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AES 128 Digital IP

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Glossary

AES	Advanced Encryption Standard
DRC	Design rules check
GF	Global Foundries
IP	Intellectual Property
IPRL	IP Readiness Level
LVS	Layout vs schematic
NIST	National Institute of Standards and Technology
PDK	Process Design Kit
RTL	Register-Transfer Logic
SPN	Substitution-Permutation Network

1 AES-128 Core IP

1.1 Overview and Origin

AES-128 (Advanced Encryption Standard) is the most popular and widely used symmetric-key block cipher algorithm standardized by National Institute of Standards and Technology (NIST). It encrypts fixed-size 128-bit data blocks using a 128-bit secret key through 10 encryption rounds. AES-128 is widely adopted in secure communication and embedded systems due to its strong security, efficiency, and suitability for both software and hardware implementations.

This IP is derived from the Apache-2.0-licensed [AES128_GFMPW0 IP](#). It provides an implementation of this algorithm in GF180mcuC Process Design Kit (PDK), performed using OpenLane tools.

The resulting package was enhanced with a reproducible physical design flow based on LibreLane, together with implementation reports and release deliverables.

The current release targets the Global Foundries (GF) 180nm CMOS process design kit using the gfl80mcu_fd_sc_mcu7t5v0 standard-cell library. In addition to the synthesizable RTL implementation, the release includes a placed-and-routed hard macro and the associated logical, physical, timing, and verification views required for integration into a larger design.

1.1.1 Functional Architecture

AES-128 is a symmetric block cipher based on a Substitution-Permutation Network (SPN). It encrypts 128-bit data blocks using a 128-bit key, representing each block as a 4×4 matrix of 16 bytes. All internal operations are performed on bytes.

The encryption process consists of an initial AddRoundKey operation followed by 10 rounds. Each round includes four transformations: SubBytes, ShiftRows, MixColumns, and AddRoundKey, except for the final round, which omits the MixColumns operation. Round keys are generated from the original key through the AES key expansion algorithm.

Figure 1 describes the typical first round of the AES encryption process.

1.2 Interfaces

The AES-128 Intellectual Property (IP) communicates with the host system through a Wishbone slave interface, which is used to configure the core, load the encryption key and plaintext data, start an

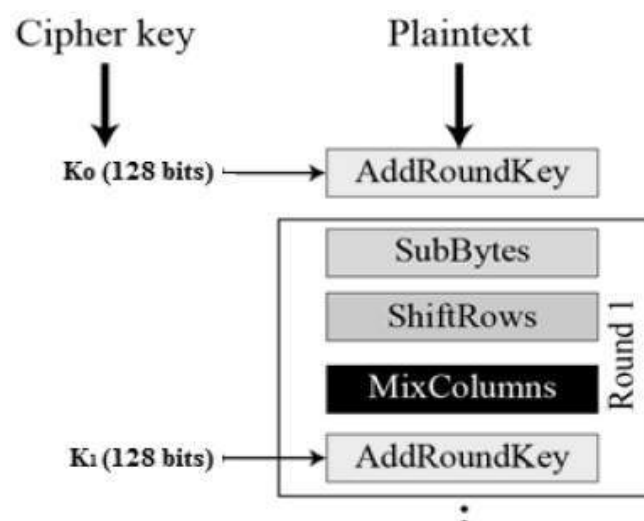


Figure 1: AES functionality scheme, from: [Source repository](#)

encryption operation, and read the resulting ciphertext. the top RTL code provides logical analyzer interface for debugging purposes and I/Os signals for integration with the surrounding SoC. that could be used by modifying the implementation. In the current implementation, these interfaces are not used by the AES core and are tied to constant values, but they can repurposed with appropriate modifications.

Table 1: Main interfaces.

Interface	Main signals	Description
Clock and reset	<code>wb_clk_i</code> , <code>wb_rst_i</code>	system clock and active-low reset.
Wishbone Slave Interface	<code>wbs_*</code>	Provides communication between the host processor and the AES core through Wishbone bus. It includes address, data, control, and handshake signals for read/write transactions.
Logic Analyzer Interface	<code>la_*</code>	Unused.
General-Purpose I/O	<code>io_*</code>	Unused.

1.3 Design and Implementation Flow

The IP was implemented using an automated RTL-to-GDS flow using LibreLane infrastructure library, covering logical verification, synthesis, physical implementation, timing analysis, physical verification, and release packaging.

The main stages were as follows:

1. RTL simulation using Icarus Verilog;
2. RTL linting using Verilator;
3. logic synthesis using Yosys;
4. initial floorplanning and power-distribution planning;
5. standard-cell placement and optimization;
6. clock-tree synthesis;
7. global and detailed routing;
8. parasitic extraction;
9. multi-corner static timing analysis;
10. design-rule checking, layout-versus-schematic verification, and antenna checking;
11. packaging of logical, timing, physical, verification, and documentation views.

The implementation environment is summarized below:

- Flow: LibreLane;
- Process design kit: `gf180mcuC`;
- Standard-cell library: `gf180mcu_fd_sc_mcu7t5v0`;
- Clock period constraint: 15 ns;
- Nominal target frequency: 67 MHz;
- Execution and testing platform: EuroCDP cloud server.

1.4 Physical Implementation Results

The principal implementation and sign-off results are reported in Table 2.

1.5 Verification Status

The verification and sign-off status of the current release is summarized in Table 3.

Table 2: Physical implementation results.

Metric	Result
Core area	1,275,180 μm^2
Clock constraint	15 ns
Setup violations	0
Hold violations	0
DRC violations	0
Antenna violations	0
LVS	Circuits match uniquely

The completed verification activities provide evidence that the Register-Transfer Logic (RTL) is functional, synthesizable, and physically implementable under the selected configuration. Gate-level simulations were completed for both the post-synthesis and post-route representations.

The remaining lint warnings do not prevent compilation or implementation, but they should be reviewed before the design is reused in a safety-critical or production context.

1.6 Maturity Assessment

The IP is assessed at **IP Readiness Level (IPRL) 5—Full design at layout level**.

The RTL has been functionally verified, synthesized, placed, clock-routed, and fully routed. The resulting hard macro has passed the available block-level static timing, Design rules check (DRC), Layout vs schematic (LVS), and antenna checks. The release also includes documentation describing the AES algorithm and its operation, although it provides limited information on the hardware implementation and interface. Nevertheless, the interface and implementation follow standard design practices; therefore, the provided information is sufficient to support evaluation and integration activities.

The IP remains pre-silicon. It has not been integrated into a fabricated test chip, characterized on silicon, or validated under physical operating conditions. It therefore does not meet the requirements of IPRL6.

Table 3: Verification status.

Verification activity	Status
RTL simulation	Passed
Synthesis checks	Passed
Post-synthesis gate-level simulation	Passed
Post-route gate-level simulation	Passed
Design-rule checking	Passed
Layout-versus-schematic verification	Passed
Multi-corner static timing analysis	Passed
Antenna verification	Passed
Lint	0 errors, 25 warnings
Clock-domain crossing analysis	Not applicable
Logical equivalence checking / EQY	Not performed
Silicon validation	Not performed

1.7 Known Limitations

The following limitations apply to the current release:

- Dedicated clock-domain crossing analysis has not been performed.
- Formal logical equivalence checking between the RTL and the synthesized or routed netlists has not been completed.
- No scan insertion, design-for-test implementation, or production test strategy is included.
- No silicon fabrication, measurement, or characterization results are available.
- The reported timing, power, and physical-verification results apply only to the recorded gfl80mcuC PDK revision, gfl80mcu_fd_sc_mcu7t5v0 library, tool configuration, operating corners, and design constraints.
- The release has been verified at block level and has not yet been validated in a complete top-level system integration.
- The original IP documentation focuses on the AES-128 algorithm, with limited information about the hardware implementation. Interface and implementation details were obtained from the RTL source code.

1.8 Conclusion

The IP provides a physically implemented and integration-documented UART hard macro targeting the gfl80mcuC technology. Its RTL, synthesis, place-and-route, gate-level simulation, static timing, DRC, LVS, and antenna results support an IPRL5 maturity assessment.